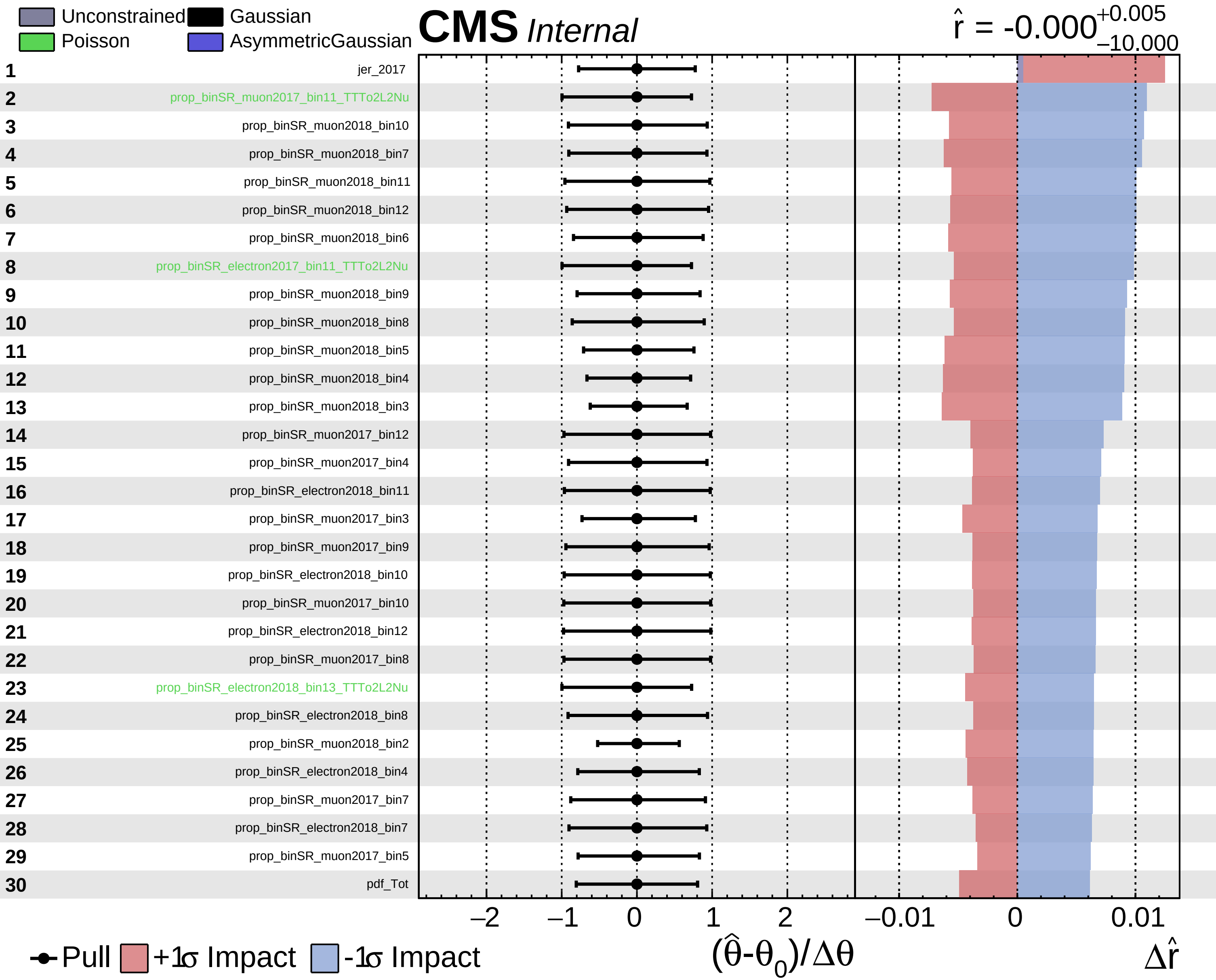
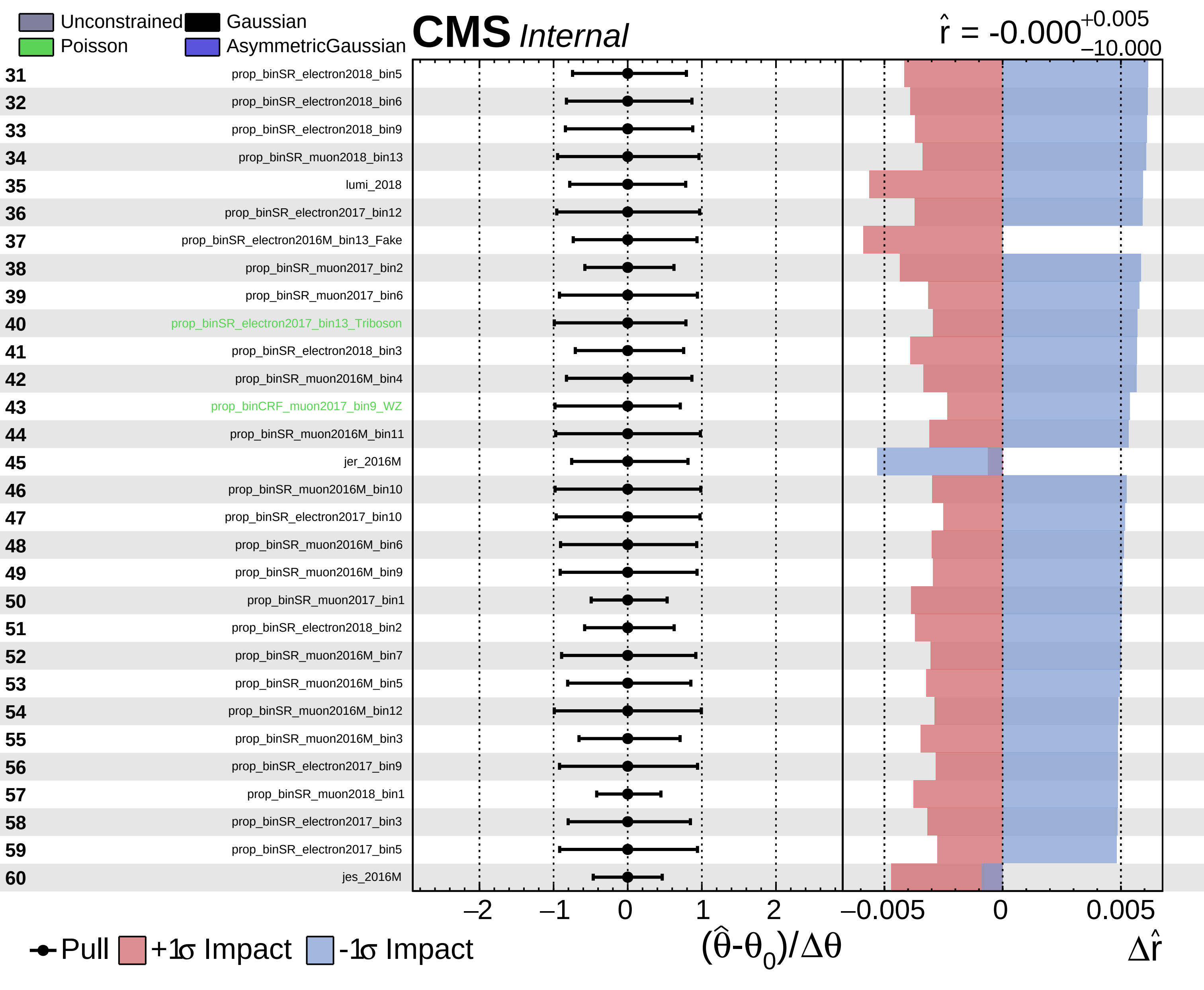
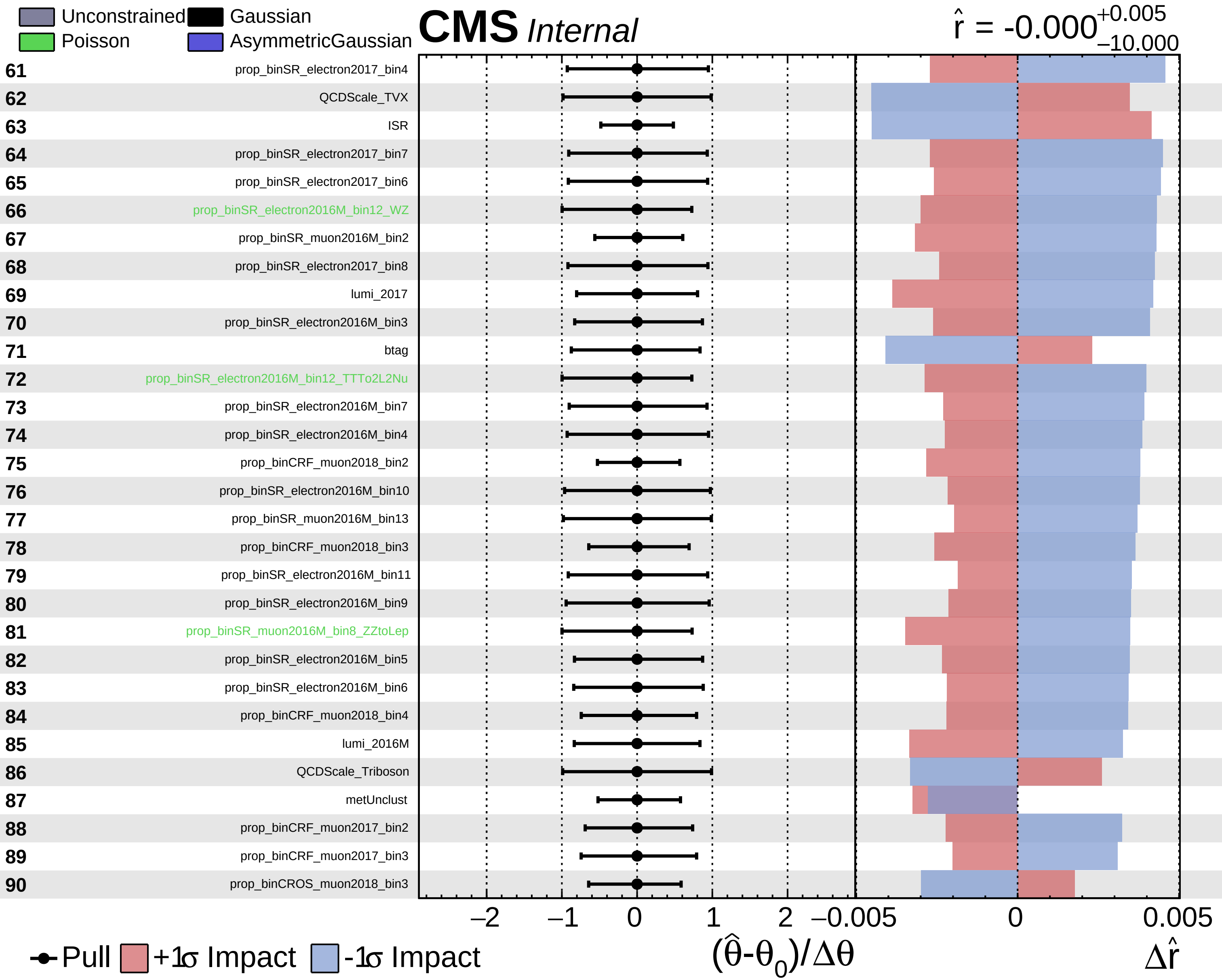
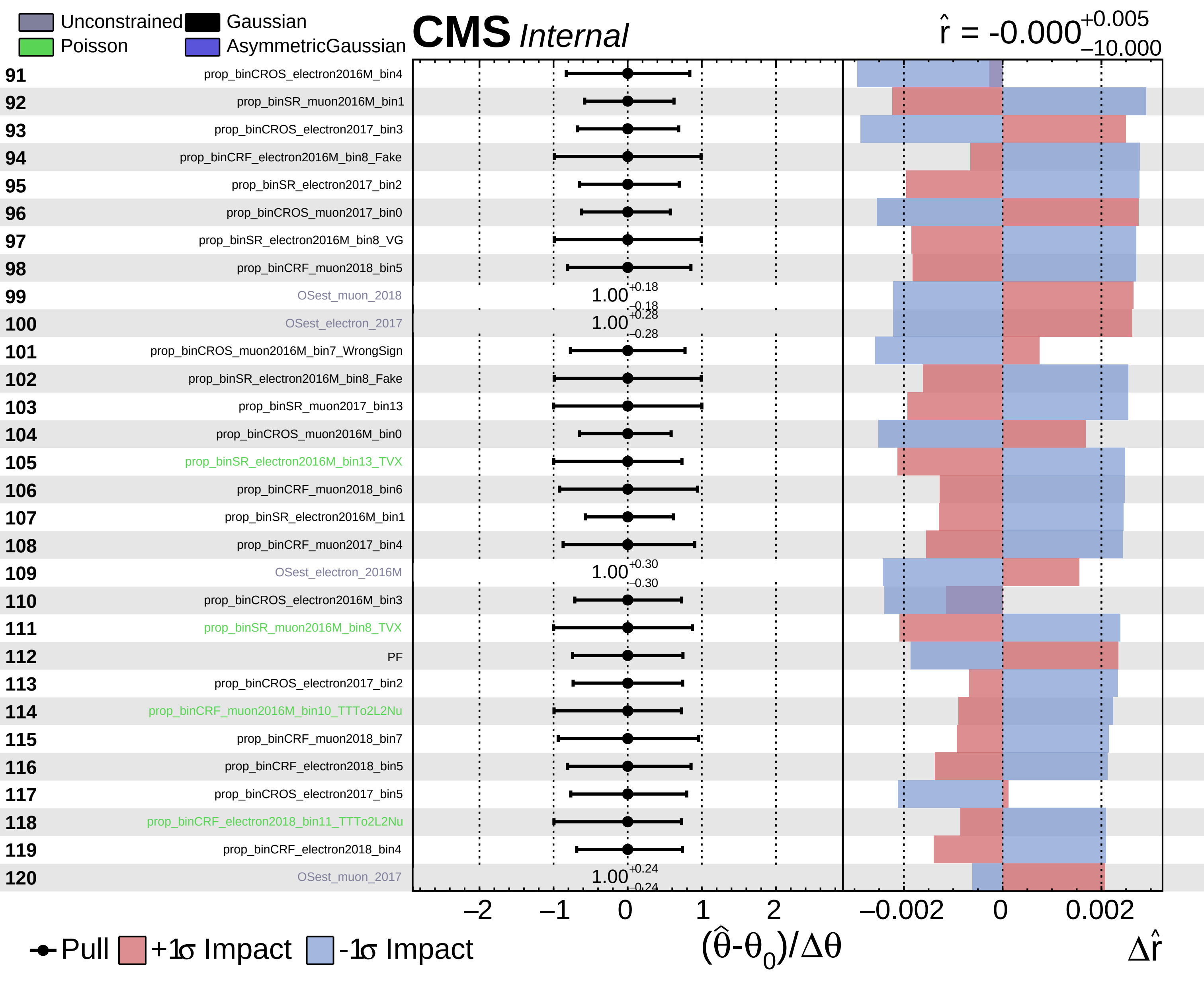


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.000^{+0.005}_{-10.000}$
1 jer_2017
2 prop_binSR_muon2017_bin11_TTTo2L2Nu
3 prop_binSR_muon2018_bin10
4 prop_binSR_muon2018_bin7
5 prop_binSR_muon2018_bin11
6 prop_binSR_muon2018_bin12
7 prop_binSR_muon2018_bin6
8 prop_binSR_electron2017_bin11_TTTo2L2Nu
9 prop_binSR_muon2018_bin9
10 prop_binSR_muon2018_bin8
11 prop_binSR_muon2018_bin5
12 prop_binSR_muon2018_bin4
13 prop_binSR_muon2018_bin3
14 prop_binSR_muon2017_bin12
15 prop_binSR_muon2017_bin4
16 prop_binSR_electron2018_bin11
17 prop_binSR_muon2017_bin3
18 prop_binSR_muon2017_bin9
19 prop_binSR_electron2018_bin10
20 prop_binSR_muon2017_bin10
21 prop_binSR_electron2018_bin12
22 prop_binSR_muon2017_bin8
23 prop_binSR_electron2018_bin13_TTTo2L2Nu
24 prop_binSR_electron2018_bin8
25 prop_binSR_muon2018_bin2
26 prop_binSR_electron2018_bin4
27 prop_binSR_muon2017_bin7
28 prop_binSR_electron2018_bin7
29 prop_binSR_muon2017_bin5
30 pdf_Tot
−2
−1
0
1
2
−0.01
0
0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

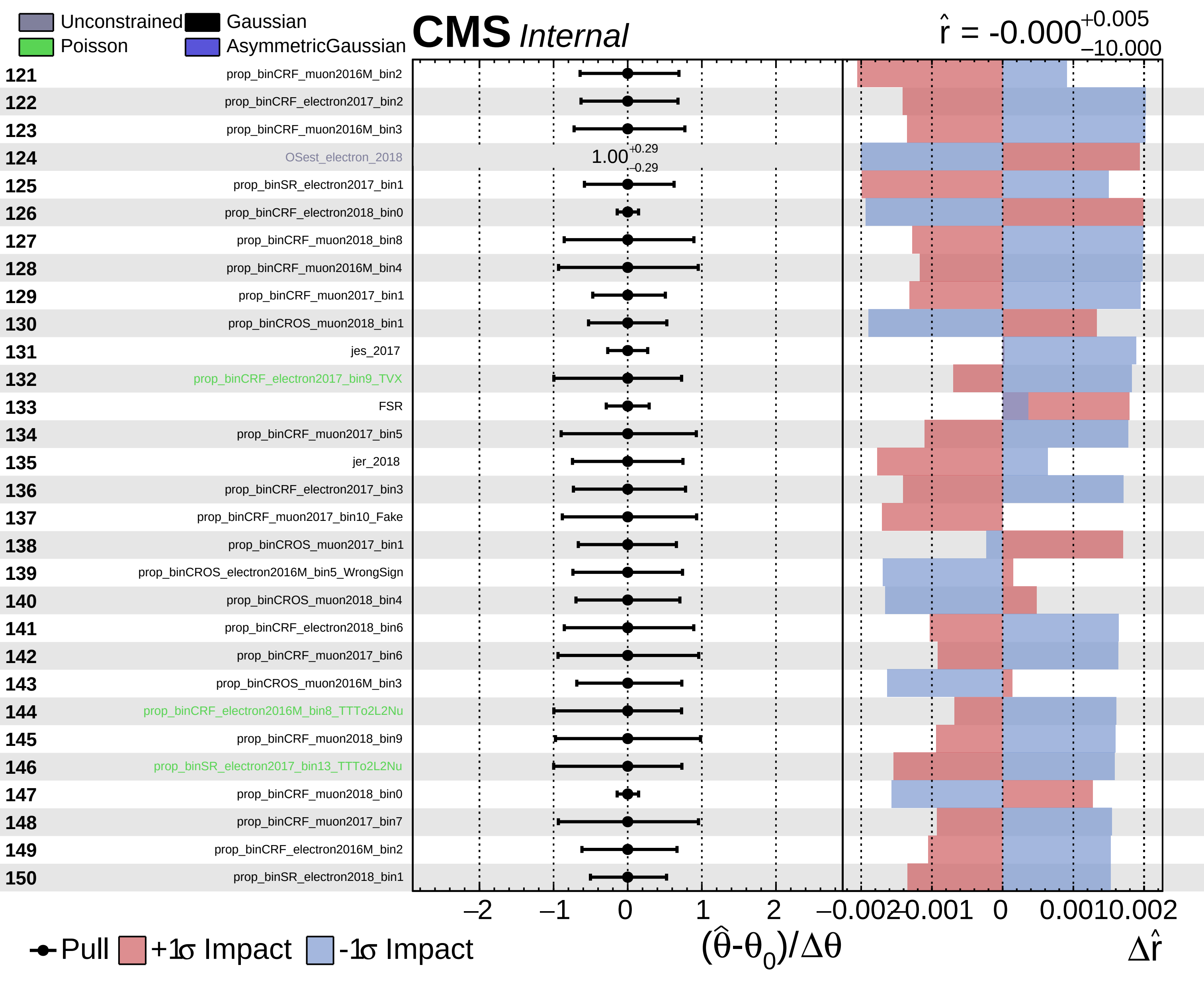

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.000^{+0.005}_{-10.000}$
31 prop_binSR_electron2018_bin5
32 prop_binSR_electron2018_bin6
33 prop_binSR_electron2018_bin9
34 prop_binSR_muon2018_bin13
35 lumi_2018
36 prop_binSR_electron2017_bin12
37 prop_binSR_electron2016M_bin13_Fake
38 prop_binSR_muon2017_bin2
39 prop_binSR_muon2017_bin6
40 prop_binSR_electron2017_bin13_Triboson
41 prop_binSR_electron2018_bin3
42 prop_binSR_muon2016M_bin4
43 prop_binCRF_muon2017_bin9_WZ
44 prop_binSR_muon2016M_bin11
45 jer_2016M
46 prop_binSR_muon2016M_bin10
47 prop_binSR_electron2017_bin10
48 prop_binSR_muon2016M_bin6
49 prop_binSR_muon2016M_bin9
50 prop_binSR_muon2017_bin1
51 prop_binSR_electron2018_bin2
52 prop_binSR_muon2016M_bin7
53 prop_binSR_muon2016M_bin5
54 prop_binSR_muon2016M_bin12
55 prop_binSR_muon2016M_bin3
56 prop_binSR_electron2017_bin9
57 prop_binSR_muon2018_bin1
58 prop_binSR_electron2017_bin3
59 prop_binSR_electron2017_bin5
60 jes_2016M
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



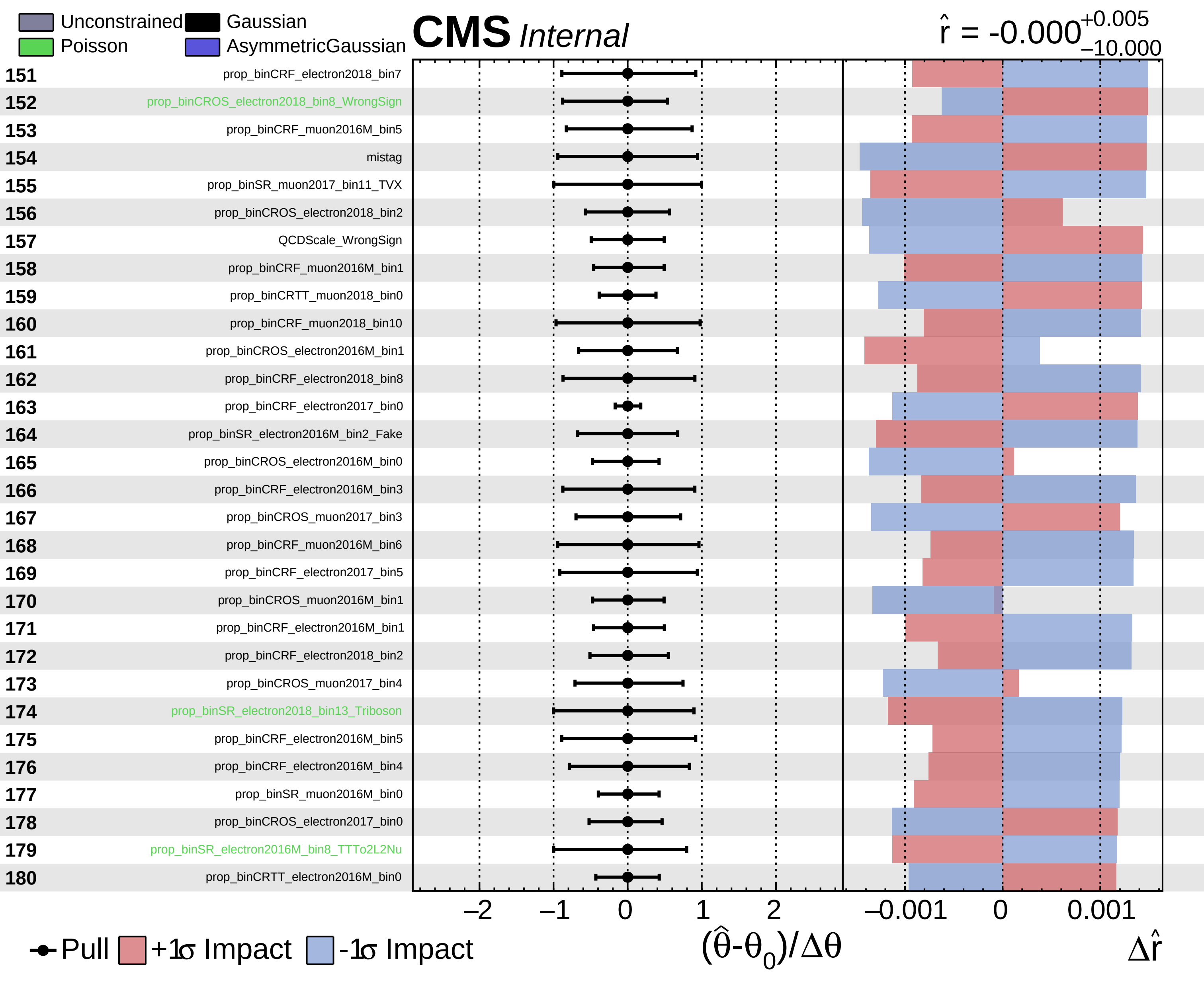
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.000^{+0.005}_{-10.000}$
61 prop_binSR_electron2017_bin4
62 QCDScale_TVX
63 ISR
64 prop_binSR_electron2017_bin7
65 prop_binSR_electron2017_bin6
66 prop_binSR_electron2016M_bin12_WZ
67 prop_binSR_muon2016M_bin2
68 prop_binSR_electron2017_bin8
69 lumi_2017
70 prop_binSR_electron2016M_bin3
71 btag
72 prop_binSR_electron2016M_bin12_TTTo2L2Nu
73 prop_binSR_electron2016M_bin7
74 prop_binSR_electron2016M_bin4
75 prop_binCRF_muon2018_bin2
76 prop_binSR_electron2016M_bin10
77 prop_binSR_muon2016M_bin13
78 prop_binCRF_muon2018_bin3
79 prop_binSR_electron2016M_bin11
80 prop_binSR_electron2016M_bin9
81 prop_binSR_muon2016M_bin8_ZZtoLep
82 prop_binSR_electron2016M_bin5
83 prop_binSR_electron2016M_bin6
84 prop_binCRF_muon2018_bin4
85 lumi_2016M
86 QCDScale_Triboson
87 metUnclust
88 prop_binCRF_muon2017_bin2
89 prop_binCRF_muon2017_bin3
90 prop_binCROS_muon2018_bin3
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

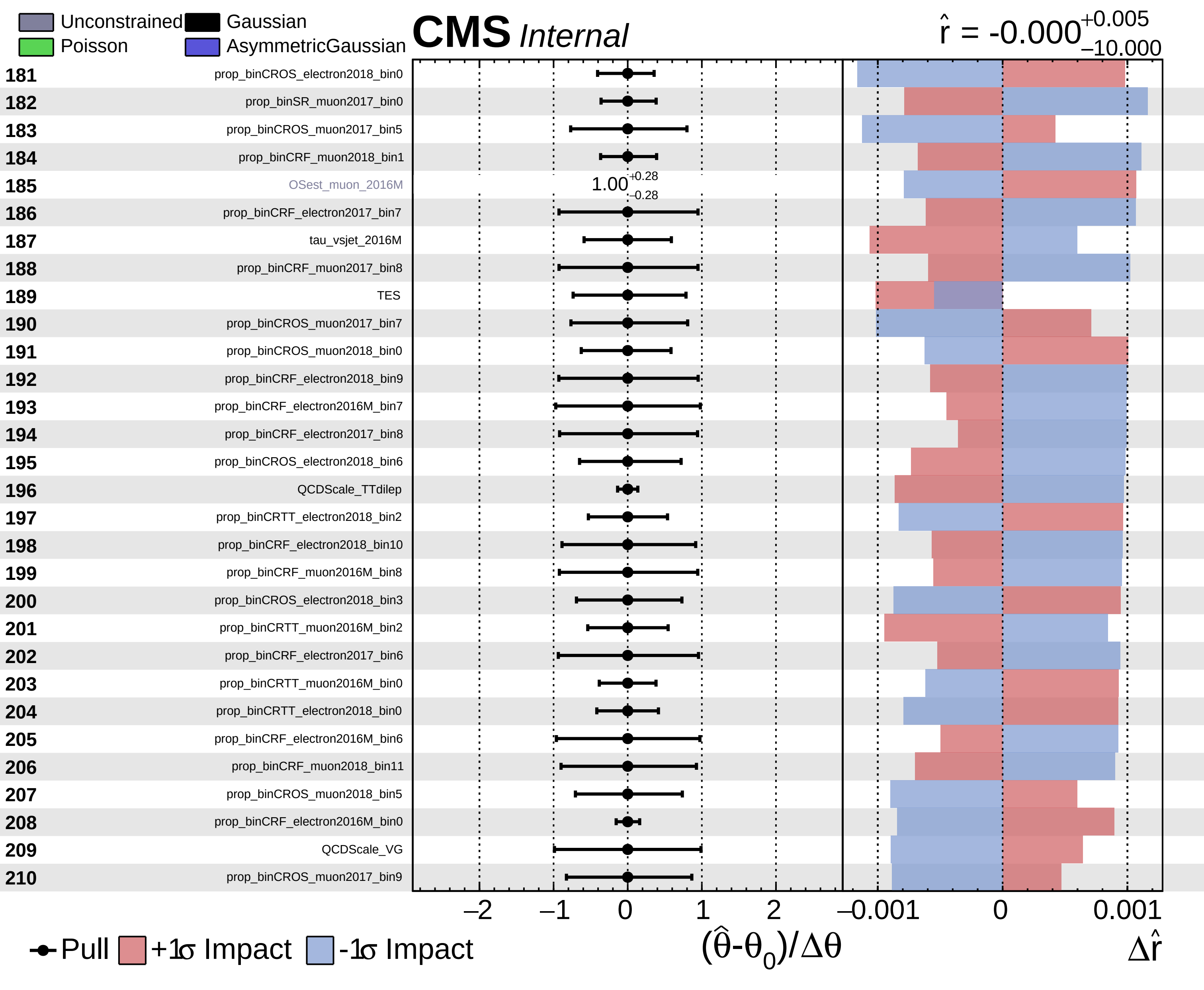


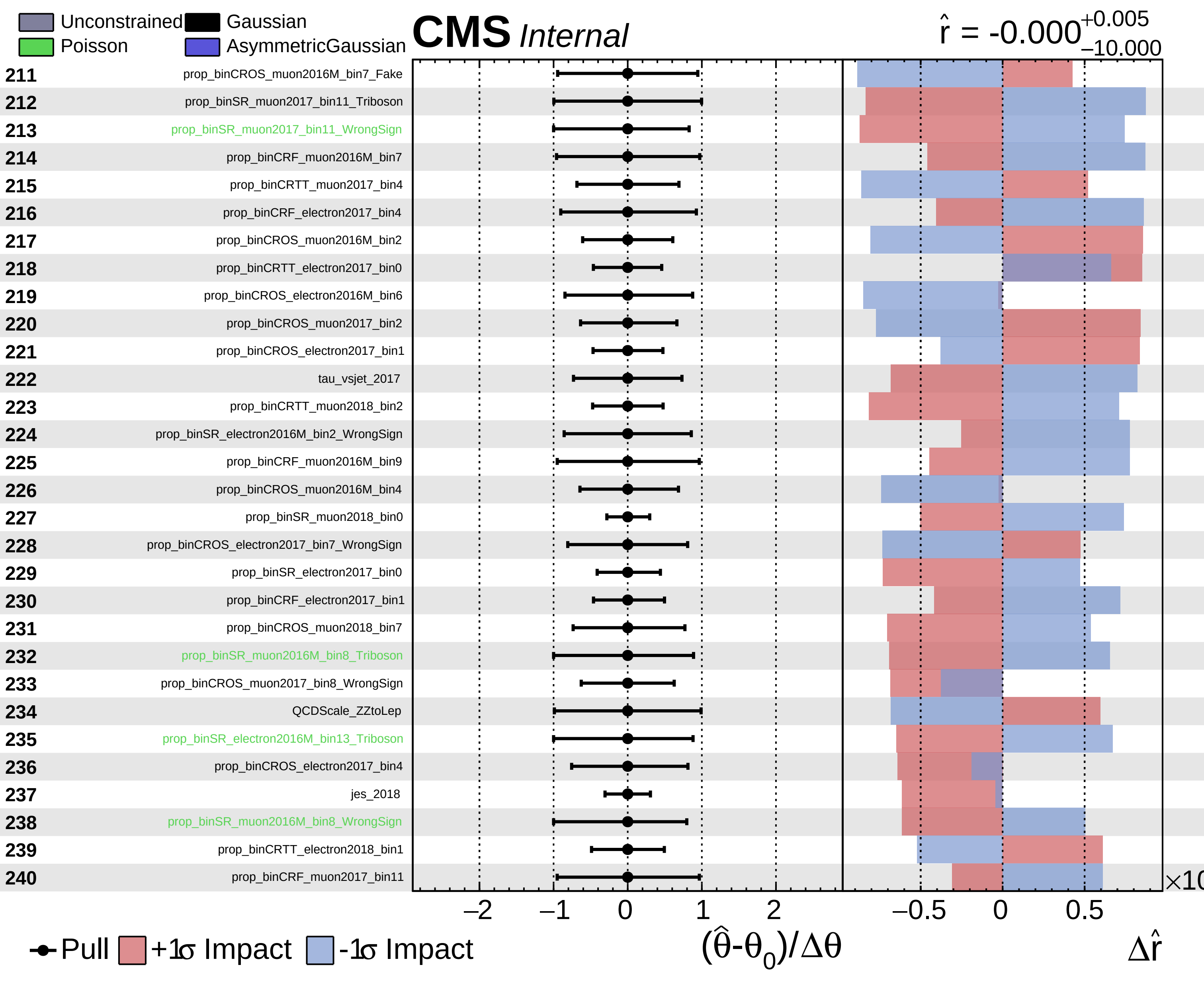
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.000^{+0.005}_{-10.000}$
91 prop_binCROS_electron2016M_bin4
92 prop_binSR_muon2016M_bin1
93 prop_binCROS_electron2017_bin3
94 prop_binCRF_electron2016M_bin8_Fake
95 prop_binSR_electron2017_bin2
96 prop_binCROS_muon2017_bin0
97 prop_binSR_electron2016M_bin8_VG
98 prop_binCRF_muon2018_bin5
99 OSest_muon_2018
$1.00^{+0.18}_{-0.18}$
100 OSest_electron_2017
$1.00^{+0.28}_{-0.28}$
101 prop_binCROS_muon2016M_bin7_WrongSign
102 prop_binSR_electron2016M_bin8_Fake
103 prop_binSR_muon2017_bin13
104 prop_binCROS_muon2016M_bin0
105 prop_binSR_electron2016M_bin13_TVX
106 prop_binCRF_muon2018_bin6
107 prop_binSR_electron2016M_bin1
108 prop_binCRF_muon2017_bin4
109 OSest_electron_2016M
$1.00^{+0.30}_{-0.30}$
110 prop_binCROS_electron2016M_bin3
111 prop_binSR_muon2016M_bin8_TVX
112 PF
113 prop_binCROS_electron2017_bin2
114 prop_binCRF_muon2016M_bin10_TTTo2L2Nu
115 prop_binCRF_muon2018_bin7
116 prop_binCRF_electron2018_bin5
117 prop_binCROS_electron2017_bin5
118 prop_binCRF_electron2018_bin11_TTTo2L2Nu
119 prop_binCRF_electron2018_bin4
120 OSest_muon_2017
$1.00^{+0.24}_{-0.24}$
−2 −1 0 1 2
−0.002 0 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



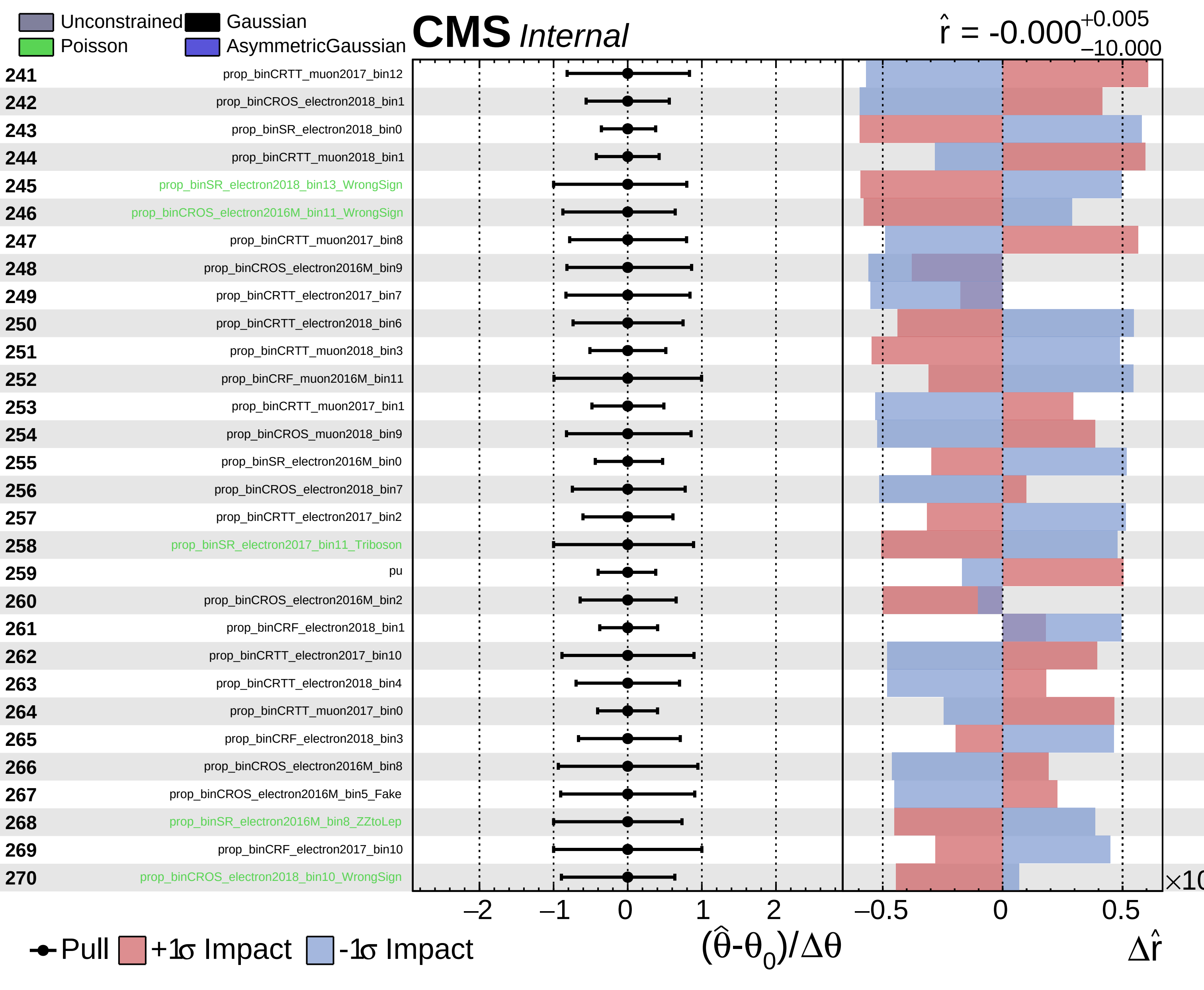
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.000^{+0.005}_{-10.000}$
121 prop_binCRF_muon2016M_bin2
122 prop_binCRF_electron2017_bin2
123 prop_binCRF_muon2016M_bin3
124 OSest_electron_2018
$1.00^{+0.29}_{-0.29}$
125 prop_binSR_electron2017_bin1
126 prop_binCRF_electron2018_bin0
127 prop_binCRF_muon2018_bin8
128 prop_binCRF_muon2016M_bin4
129 prop_binCRF_muon2017_bin1
130 prop_binCROS_muon2018_bin1
131 jes_2017
132 prop_binCRF_electron2017_bin9_TVX
133 FSR
134 prop_binCRF_muon2017_bin5
135 jer_2018
136 prop_binCRF_electron2017_bin3
137 prop_binCRF_muon2017_bin10_Fake
138 prop_binCROS_muon2017_bin1
139 prop_binCROS_electron2016M_bin5_WrongSign
140 prop_binCROS_muon2018_bin4
141 prop_binCRF_electron2018_bin6
142 prop_binCRF_muon2017_bin6
143 prop_binCROS_muon2016M_bin3
144 prop_binCRF_electron2016M_bin8_TTTo2L2Nu
145 prop_binCRF_muon2018_bin9
146 prop_binSR_electron2017_bin13_TTTo2L2Nu
147 prop_binCRF_muon2018_bin0
148 prop_binCRF_muon2017_bin7
149 prop_binCRF_electron2016M_bin2
150 prop_binSR_electron2018_bin1
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.000^{+0.005}_{-10.000}$
151 prop_binCRF_electron2018_bin7
152 prop_binCROS_electron2018_bin8_WrongSign
153 prop_binCRF_muon2016M_bin5
154 mistag
155 prop_binSR_muon2017_bin11_TVX
156 prop_binCROS_electron2018_bin2
157 QCDScale_WrongSign
158 prop_binCRF_muon2016M_bin1
159 prop_binCRTT_muon2018_bin0
160 prop_binCRF_muon2018_bin10
161 prop_binCROS_electron2016M_bin1
162 prop_binCRF_electron2018_bin8
163 prop_binCRF_electron2017_bin0
164 prop_binSR_electron2016M_bin2_Fake
165 prop_binCROS_electron2016M_bin0
166 prop_binCRF_electron2016M_bin3
167 prop_binCROS_muon2017_bin3
168 prop_binCRF_muon2016M_bin6
169 prop_binCRF_electron2017_bin5
170 prop_binCROS_muon2016M_bin1
171 prop_binCRF_electron2016M_bin1
172 prop_binCRF_electron2018_bin2
173 prop_binCROS_muon2017_bin4
174 prop_binSR_electron2018_bin13_Triboson
175 prop_binCRF_electron2016M_bin5
176 prop_binCRF_electron2016M_bin4
177 prop_binSR_muon2016M_bin0
178 prop_binCROS_electron2017_bin0
179 prop_binSR_electron2016M_bin8_TTTo2L2Nu
180 prop_binCRTT_electron2016M_bin0
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

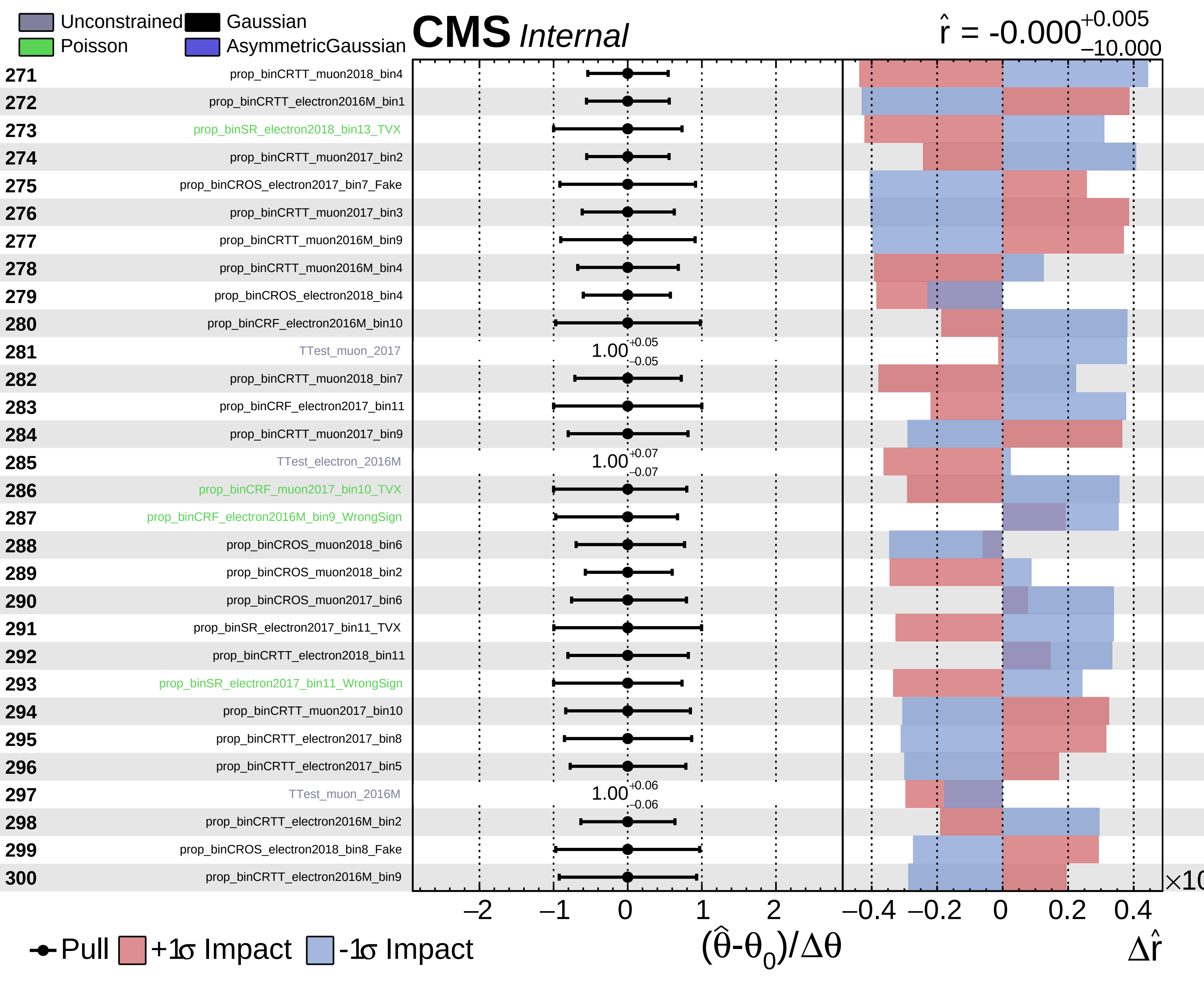

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.000^{+0.005}_{-10.000}$
181 prop_binCROS_electron2018_bin0
182 prop_binSR_muon2017_bin0
183 prop_binCROS_muon2017_bin5
184 prop_binCRF_muon2018_bin1
185 OSest_muon_2016M
$1.00^{+0.28}_{-0.28}$
186 prop_binCRF_electron2017_bin7
187 tau_vsjet_2016M
188 prop_binCRF_muon2017_bin8
189 TES
190 prop_binCROS_muon2017_bin7
191 prop_binCROS_muon2018_bin0
192 prop_binCRF_electron2018_bin9
193 prop_binCRF_electron2016M_bin7
194 prop_binCRF_electron2017_bin8
195 prop_binCROS_electron2018_bin6
196 QCDScale_TTdilep
197 prop_binCRTT_electron2018_bin2
198 prop_binCRF_electron2018_bin10
199 prop_binCRF_muon2016M_bin8
200 prop_binCROS_electron2018_bin3
201 prop_binCRTT_muon2016M_bin2
202 prop_binCRF_electron2017_bin6
203 prop_binCRTT_muon2016M_bin0
204 prop_binCRTT_electron2018_bin0
205 prop_binCRF_electron2016M_bin6
206 prop_binCRF_muon2018_bin11
207 prop_binCROS_muon2018_bin5
208 prop_binCRF_electron2016M_bin0
209 QCDScale_VG
210 prop_binCROS_muon2017_bin9
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



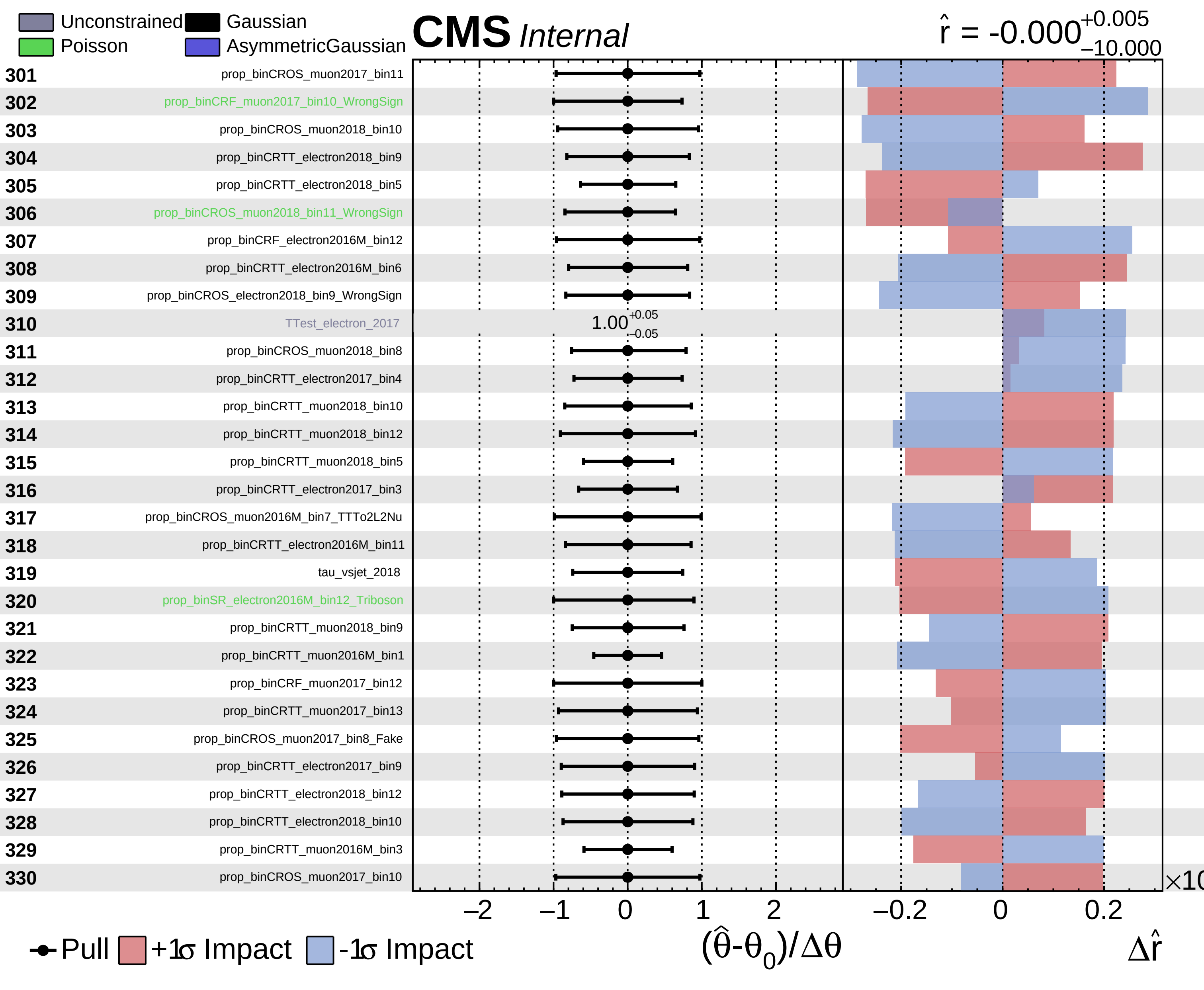
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.000^{+0.005}_{-10.000}$
211 prop_binCROS_muon2016M_bin7_Fake
212 prop_binSR_muon2017_bin11_Triboson
213 prop_binSR_muon2017_bin11_WrongSign
214 prop_binCRF_muon2016M_bin7
215 prop_binCRTT_muon2017_bin4
216 prop_binCRF_electron2017_bin4
217 prop_binCROS_muon2016M_bin2
218 prop_binCRTT_electron2017_bin0
219 prop_binCROS_electron2016M_bin6
220 prop_binCROS_muon2017_bin2
221 prop_binCROS_electron2017_bin1
222 tau_vsjet_2017
223 prop_binCRTT_muon2018_bin2
224 prop_binSR_electron2016M_bin2_WrongSign
225 prop_binCRF_muon2016M_bin9
226 prop_binCROS_muon2016M_bin4
227 prop_binSR_muon2018_bin0
228 prop_binCROS_electron2017_bin7_WrongSign
229 prop_binSR_electron2017_bin0
230 prop_binCRF_electron2017_bin1
231 prop_binCROS_muon2018_bin7
232 prop_binSR_muon2016M_bin8_Triboson
233 prop_binCROS_muon2017_bin8_WrongSign
234 QCDScale_ZZtoLep
235 prop_binSR_electron2016M_bin13_Triboson
236 prop_binCROS_electron2017_bin4
237 jes_2018
238 prop_binSR_muon2016M_bin8_WrongSign
239 prop_binCRTT_electron2018_bin1
240 prop_binCRF_muon2017_bin11
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



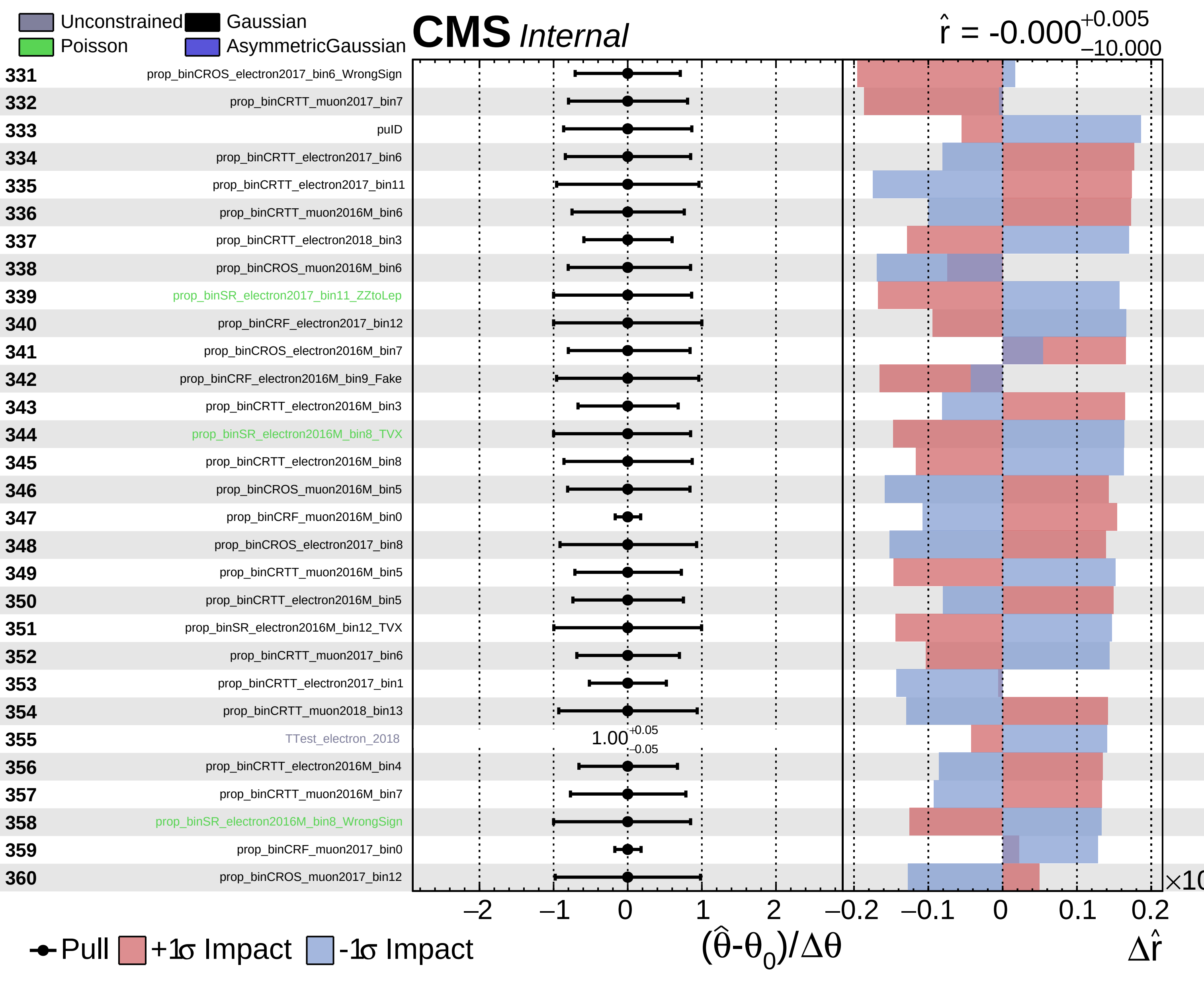
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.000^{+0.005}_{-10.000}$
241 prop_binCRTT_muon2017_bin12
242 prop_binCROS_electron2018_bin1
243 prop_binSR_electron2018_bin0
244 prop_binCRTT_muon2018_bin1
245 prop_binSR_electron2018_bin13_WrongSign
246 prop_binCROS_electron2016M_bin11_WrongSign
247 prop_binCRTT_muon2017_bin8
248 prop_binCROS_electron2016M_bin9
249 prop_binCRTT_electron2017_bin7
250 prop_binCRTT_electron2018_bin6
251 prop_binCRTT_muon2018_bin3
252 prop_binCRF_muon2016M_bin11
253 prop_binCRTT_muon2017_bin1
254 prop_binCROS_muon2018_bin9
255 prop_binSR_electron2016M_bin0
256 prop_binCROS_electron2018_bin7
257 prop_binCRTT_electron2017_bin2
258 prop_binSR_electron2017_bin11_Triboson
259 pu
260 prop_binCROS_electron2016M_bin2
261 prop_binCRF_electron2018_bin1
262 prop_binCRTT_electron2017_bin10
263 prop_binCRTT_electron2018_bin4
264 prop_binCRTT_muon2017_bin0
265 prop_binCRF_electron2018_bin3
266 prop_binCROS_electron2016M_bin8
267 prop_binCROS_electron2016M_bin5_Fake
268 prop_binSR_electron2016M_bin8_ZZtoLep
269 prop_binCRF_electron2017_bin10
270 prop_binCROS_electron2018_bin10_WrongSign
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



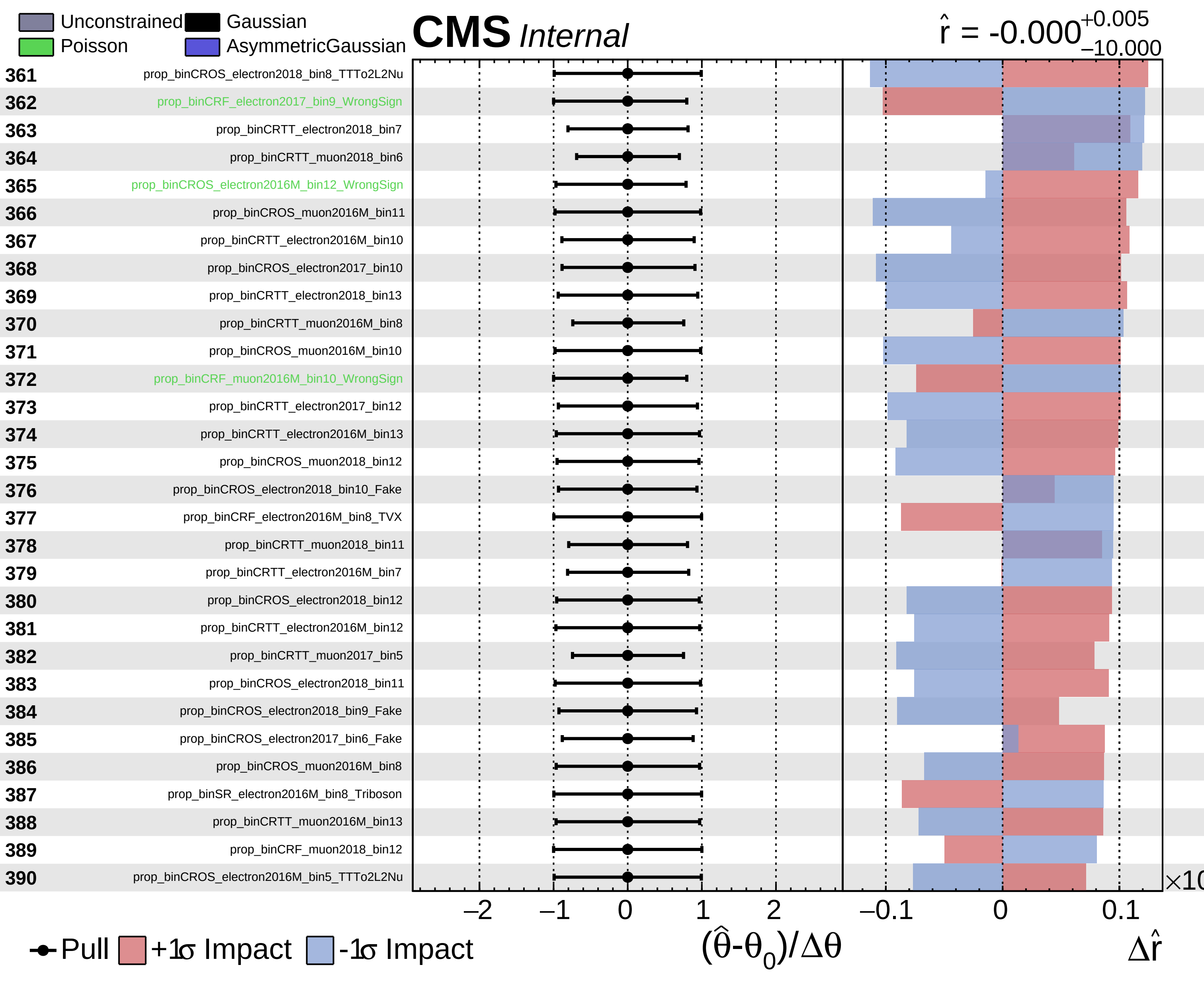
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.000^{+0.005}_{-10.000}$
271 prop_binCRTT_muon2018_bin4
272 prop_binCRTT_electron2016M_bin1
273 prop_binSR_electron2018_bin13_TVX
274 prop_binCRTT_muon2017_bin2
275 prop_binCROS_electron2017_bin7_Fake
276 prop_binCRTT_muon2017_bin3
277 prop_binCRTT_muon2016M_bin9
278 prop_binCRTT_muon2016M_bin4
279 prop_binCROS_electron2018_bin4
280 prop_binCRF_electron2016M_bin10
281 TTest_muon_2017
$1.00^{+0.05}_{-0.05}$
282 prop_binCRTT_muon2018_bin7
283 prop_binCRF_electron2017_bin11
284 prop_binCRTT_muon2017_bin9
285 TTest_electron_2016M
$1.00^{+0.07}_{-0.07}$
286 prop_binCRF_muon2017_bin10_TVX
287 prop_binCRF_electron2016M_bin9_WrongSign
288 prop_binCROS_muon2018_bin6
289 prop_binCROS_muon2018_bin2
290 prop_binCROS_muon2017_bin6
291 prop_binSR_electron2017_bin11_TVX
292 prop_binCRTT_electron2018_bin11
293 prop_binSR_electron2017_bin11_WrongSign
294 prop_binCRTT_muon2017_bin10
295 prop_binCRTT_electron2017_bin8
296 prop_binCRTT_electron2017_bin5
297 TTest_muon_2016M
$1.00^{+0.06}_{-0.06}$
298 prop_binCRTT_electron2016M_bin2
299 prop_binCROS_electron2018_bin8_Fake
300 prop_binCRTT_electron2016M_bin9
−2 −1 0 1 2
−0.4 −0.2 0 0.2 0.4
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



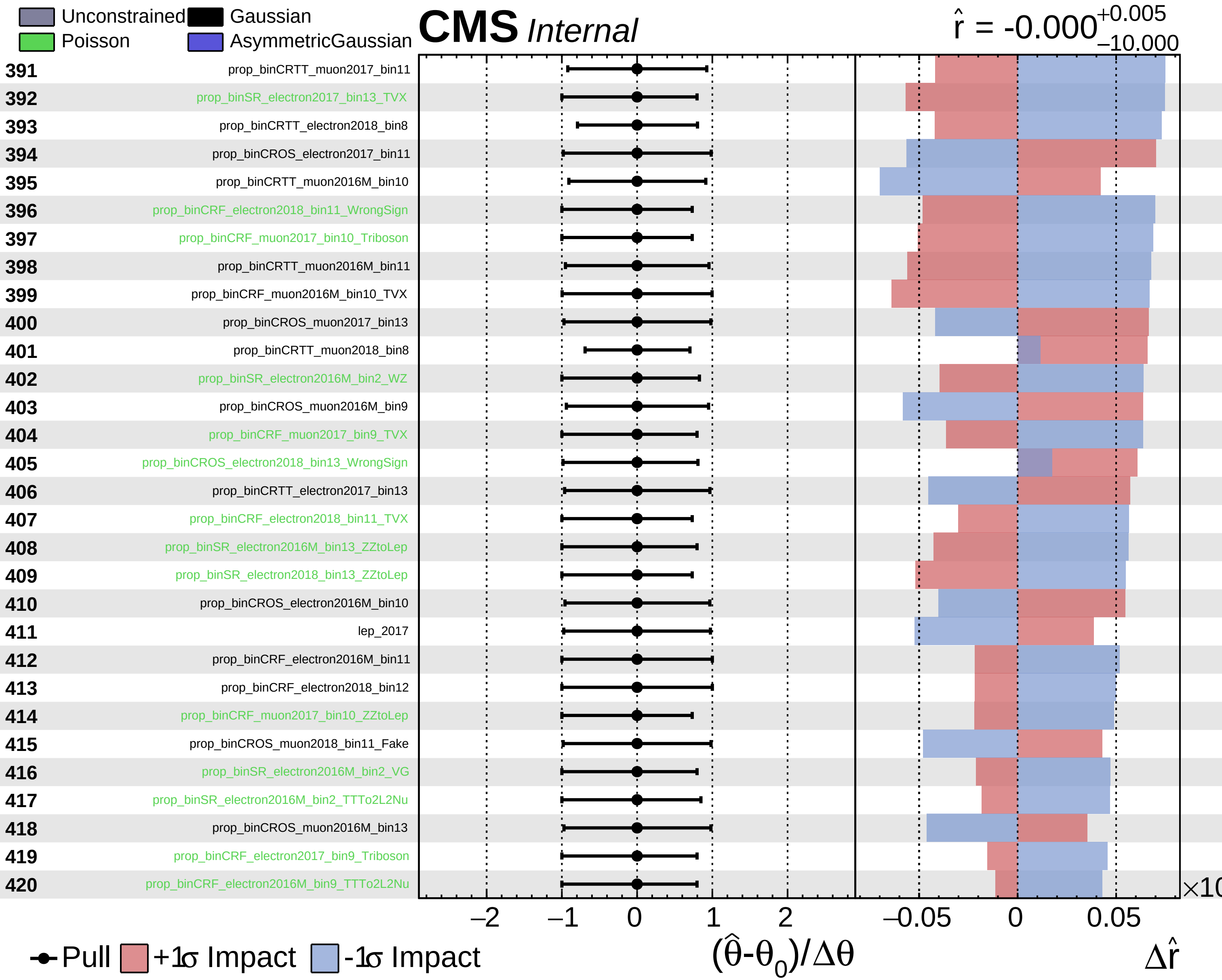
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.000^{+0.005}_{-10.000}$
301 prop_binCROS_muon2017_bin11
302 prop_binCRF_muon2017_bin10_WrongSign
303 prop_binCROS_muon2018_bin10
304 prop_binCRTT_electron2018_bin9
305 prop_binCRTT_electron2018_bin5
306 prop_binCROS_muon2018_bin11_WrongSign
307 prop_binCRF_electron2016M_bin12
308 prop_binCRTT_electron2016M_bin6
309 prop_binCROS_electron2018_bin9_WrongSign
310 TTest_electron_2017
$1.00^{+0.05}_{-0.05}$
311 prop_binCROS_muon2018_bin8
312 prop_binCRTT_electron2017_bin4
313 prop_binCRTT_muon2018_bin10
314 prop_binCRTT_muon2018_bin12
315 prop_binCRTT_muon2018_bin5
316 prop_binCRTT_electron2017_bin3
317 prop_binCROS_muon2016M_bin7_TTTo2L2Nu
318 prop_binCRTT_electron2016M_bin11
319 tau_vsjet_2018
320 prop_binSR_electron2016M_bin12_Triboson
321 prop_binCRTT_muon2018_bin9
322 prop_binCRTT_muon2016M_bin1
323 prop_binCRF_muon2017_bin12
324 prop_binCRTT_muon2017_bin13
325 prop_binCROS_muon2017_bin8_Fake
326 prop_binCRTT_electron2017_bin9
327 prop_binCRTT_electron2018_bin12
328 prop_binCRTT_electron2018_bin10
329 prop_binCRTT_muon2016M_bin3
330 prop_binCROS_muon2017_bin10
−2 −1 0 1 2
−0.2 0 0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

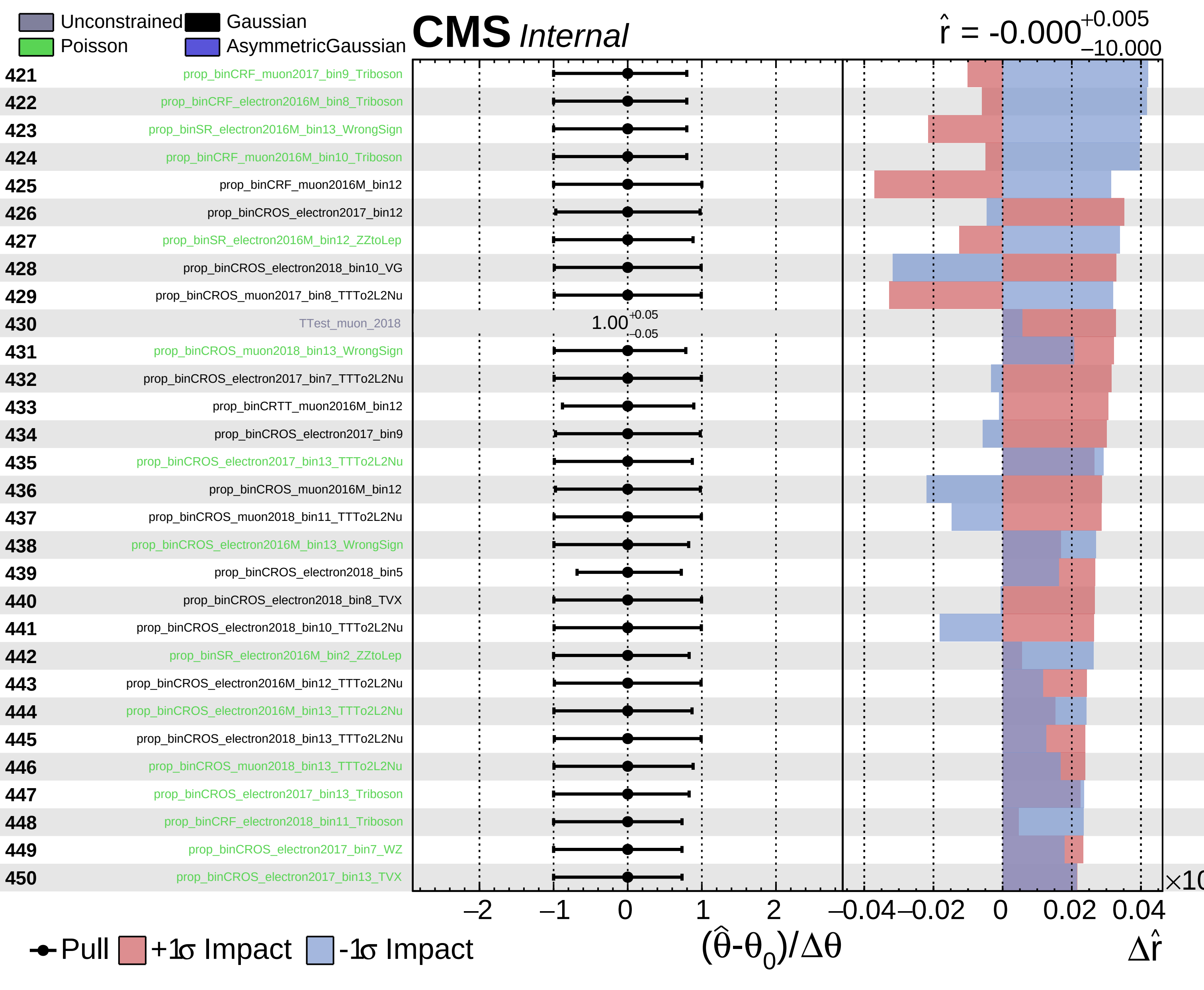


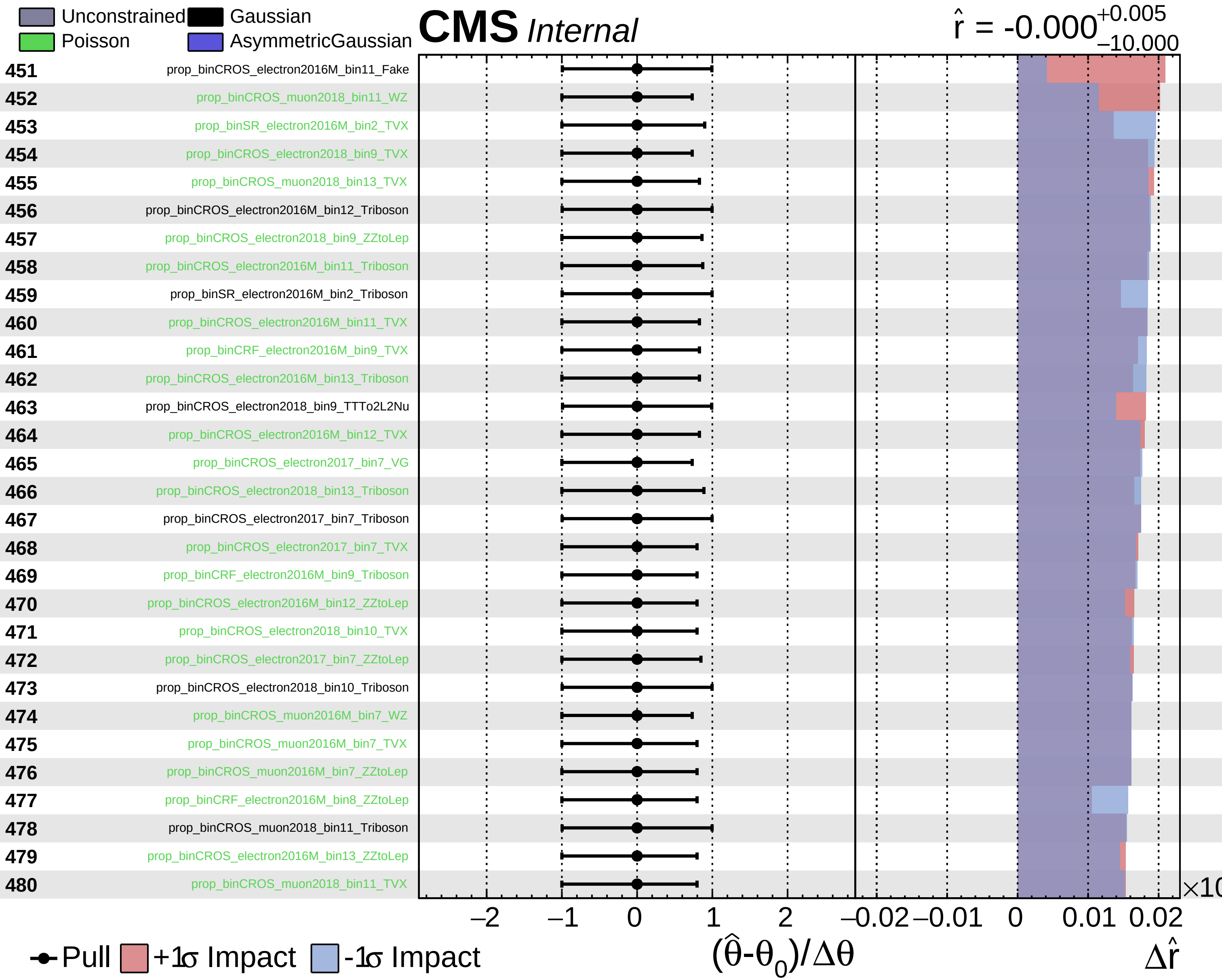
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.000 +0.005 −10.000
331 prop_binCROS_electron2017_bin6_WrongSign
332 prop_binCRTT_muon2017_bin7
333 puID
334 prop_binCRTT_electron2017_bin6
335 prop_binCRTT_electron2017_bin11
336 prop_binCRTT_muon2016M_bin6
337 prop_binCRTT_electron2018_bin3
338 prop_binCROS_muon2016M_bin6
339 prop_binSR_electron2017_bin11_ZZtoLep
340 prop_binCRF_electron2017_bin12
341 prop_binCROS_electron2016M_bin7
342 prop_binCRF_electron2016M_bin9_Fake
343 prop_binCRTT_electron2016M_bin3
344 prop_binSR_electron2016M_bin8_TVX
345 prop_binCRTT_electron2016M_bin8
346 prop_binCROS_muon2016M_bin5
347 prop_binCRF_muon2016M_bin0
348 prop_binCROS_electron2017_bin8
349 prop_binCRTT_muon2016M_bin5
350 prop_binCRTT_electron2016M_bin5
351 prop_binSR_electron2016M_bin12_TVX
352 prop_binCRTT_muon2017_bin6
353 prop_binCRTT_electron2017_bin1
354 prop_binCRTT_muon2018_bin13
355 TTest_electron_2018
1.00 +0.05 −0.05
356 prop_binCRTT_electron2016M_bin4
357 prop_binCRTT_muon2016M_bin7
358 prop_binSR_electron2016M_bin8_WrongSign
359 prop_binCRF_muon2017_bin0
360 prop_binCROS_muon2017_bin12
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



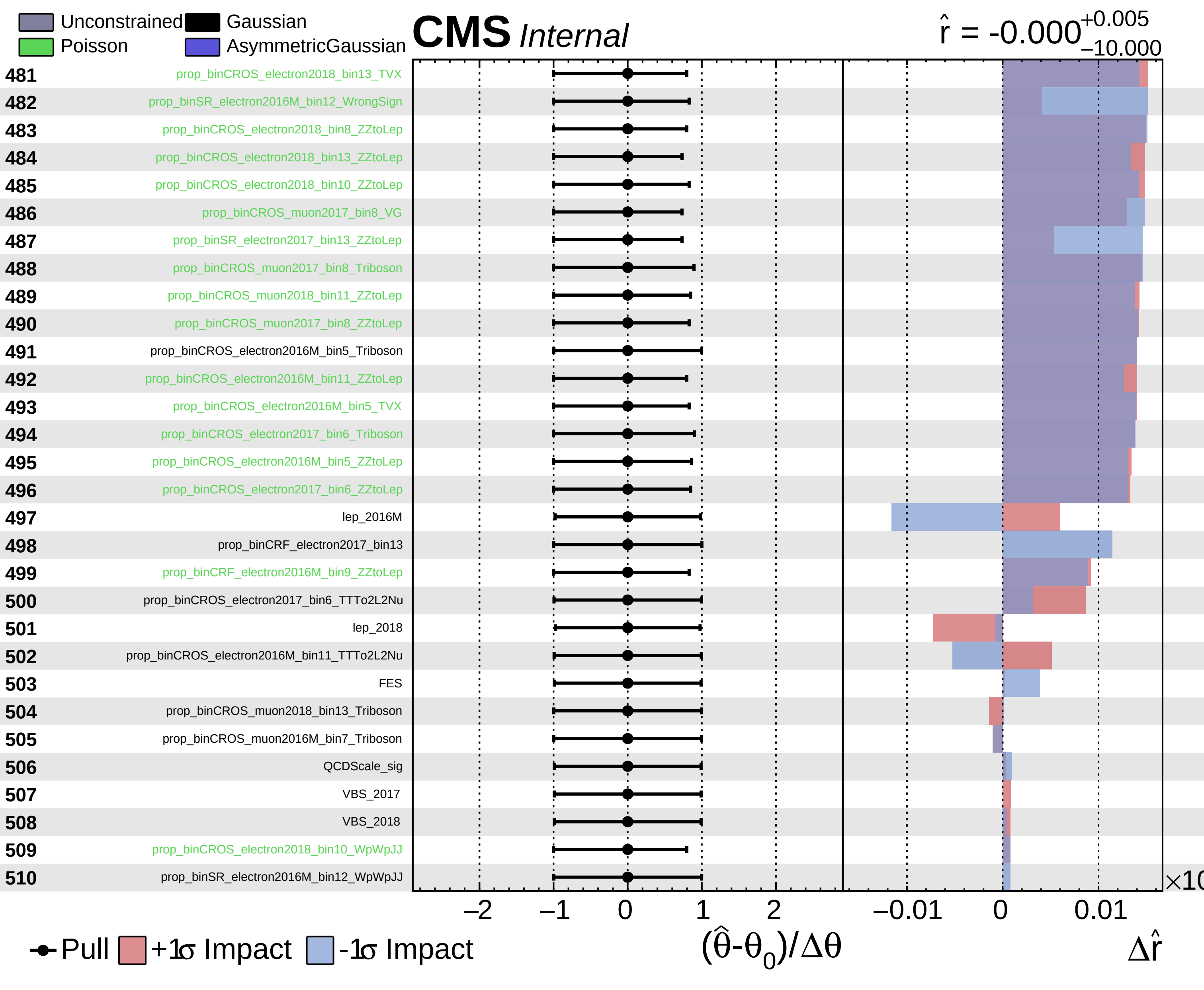
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.000^{+0.005}_{-10.000}$
361 prop_binCROS_electron2018_bin8_TTTo2L2Nu
362 prop_binCRF_electron2017_bin9_WrongSign
363 prop_binCRTT_electron2018_bin7
364 prop_binCRTT_muon2018_bin6
365 prop_binCROS_electron2016M_bin12_WrongSign
366 prop_binCROS_muon2016M_bin11
367 prop_binCRTT_electron2016M_bin10
368 prop_binCROS_electron2017_bin10
369 prop_binCRTT_electron2018_bin13
370 prop_binCRTT_muon2016M_bin8
371 prop_binCROS_muon2016M_bin10
372 prop_binCRF_muon2016M_bin10_WrongSign
373 prop_binCRTT_electron2017_bin12
374 prop_binCRTT_electron2016M_bin13
375 prop_binCROS_muon2018_bin12
376 prop_binCROS_electron2018_bin10_Fake
377 prop_binCRF_electron2016M_bin8_TVX
378 prop_binCRTT_muon2018_bin11
379 prop_binCRTT_electron2016M_bin7
380 prop_binCROS_electron2018_bin12
381 prop_binCRTT_electron2016M_bin12
382 prop_binCRTT_muon2017_bin5
383 prop_binCROS_electron2018_bin11
384 prop_binCROS_electron2018_bin9_Fake
385 prop_binCROS_electron2017_bin6_Fake
386 prop_binCROS_muon2016M_bin8
387 prop_binSR_electron2016M_bin8_Triboson
388 prop_binCRTT_muon2016M_bin13
389 prop_binCRF_muon2018_bin12
390 prop_binCROS_electron2016M_bin5_TTTo2L2Nu
−2
−1
0
1
2
−0.1
0
0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

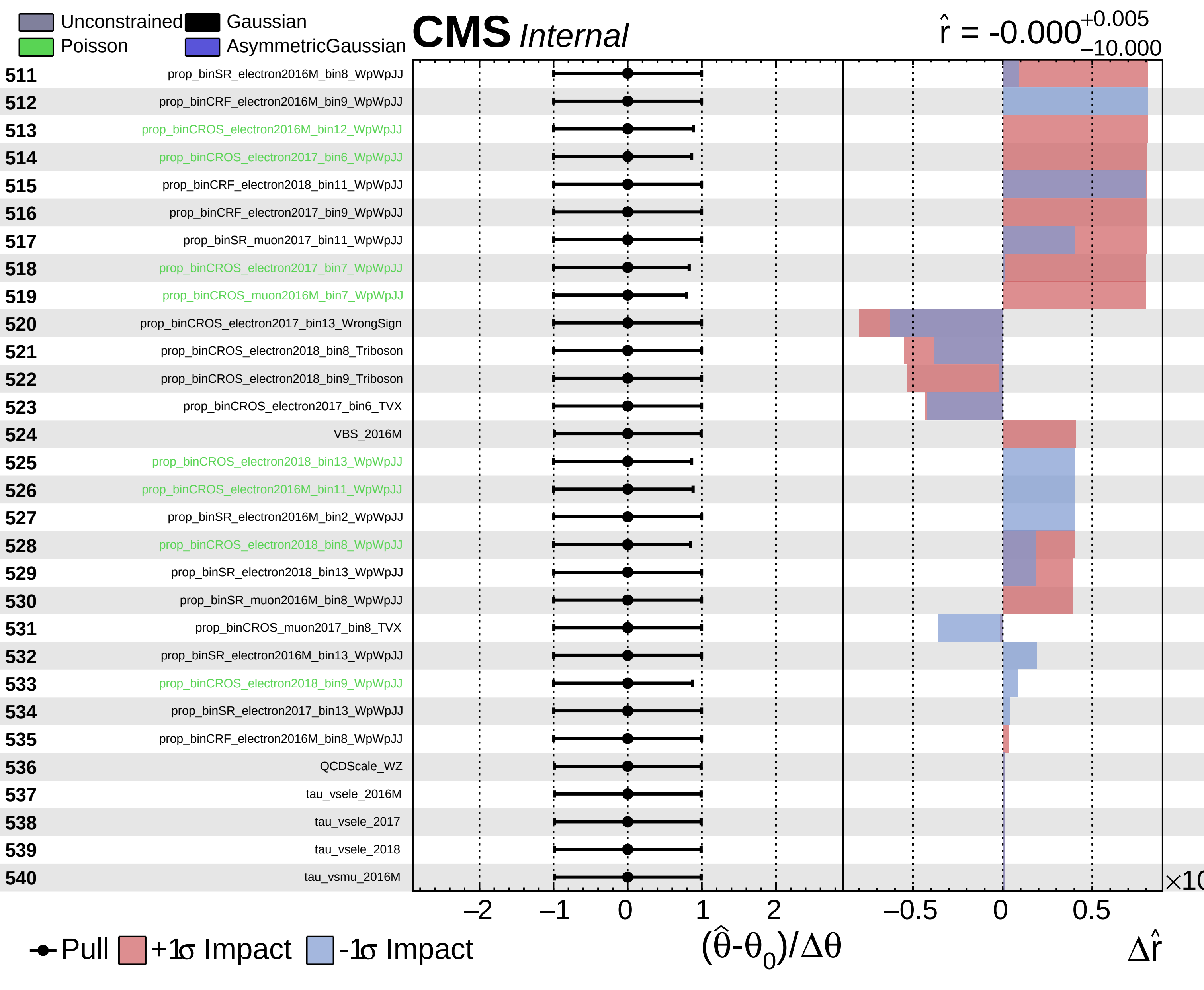

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.000 +0.005 −10.000
391 prop_binCRTT_muon2017_bin11
392 prop_binSR_electron2017_bin13_TVX
393 prop_binCRTT_electron2018_bin8
394 prop_binCROS_electron2017_bin11
395 prop_binCRTT_muon2016M_bin10
396 prop_binCRF_electron2018_bin11_WrongSign
397 prop_binCRF_muon2017_bin10_Triboson
398 prop_binCRTT_muon2016M_bin11
399 prop_binCRF_muon2016M_bin10_TVX
400 prop_binCROS_muon2017_bin13
401 prop_binCRTT_muon2018_bin8
402 prop_binSR_electron2016M_bin2_WZ
403 prop_binCROS_muon2016M_bin9
404 prop_binCRF_muon2017_bin9_TVX
405 prop_binCROS_electron2018_bin13_WrongSign
406 prop_binCRTT_electron2017_bin13
407 prop_binCRF_electron2018_bin11_TVX
408 prop_binSR_electron2016M_bin13_ZZtoLep
409 prop_binSR_electron2018_bin13_ZZtoLep
410 prop_binCROS_electron2016M_bin10
411 lep_2017
412 prop_binCRF_electron2016M_bin11
413 prop_binCRF_electron2018_bin12
414 prop_binCRF_muon2017_bin10_ZZtoLep
415 prop_binCROS_muon2018_bin11_Fake
416 prop_binSR_electron2016M_bin2_VG
417 prop_binSR_electron2016M_bin2_TTTo2L2Nu
418 prop_binCROS_muon2016M_bin13
419 prop_binCRF_electron2017_bin9_Triboson
420 prop_binCRF_electron2016M_bin9_TTTo2L2Nu
−2 −1 0 1 2
−0.05 0 0.05
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

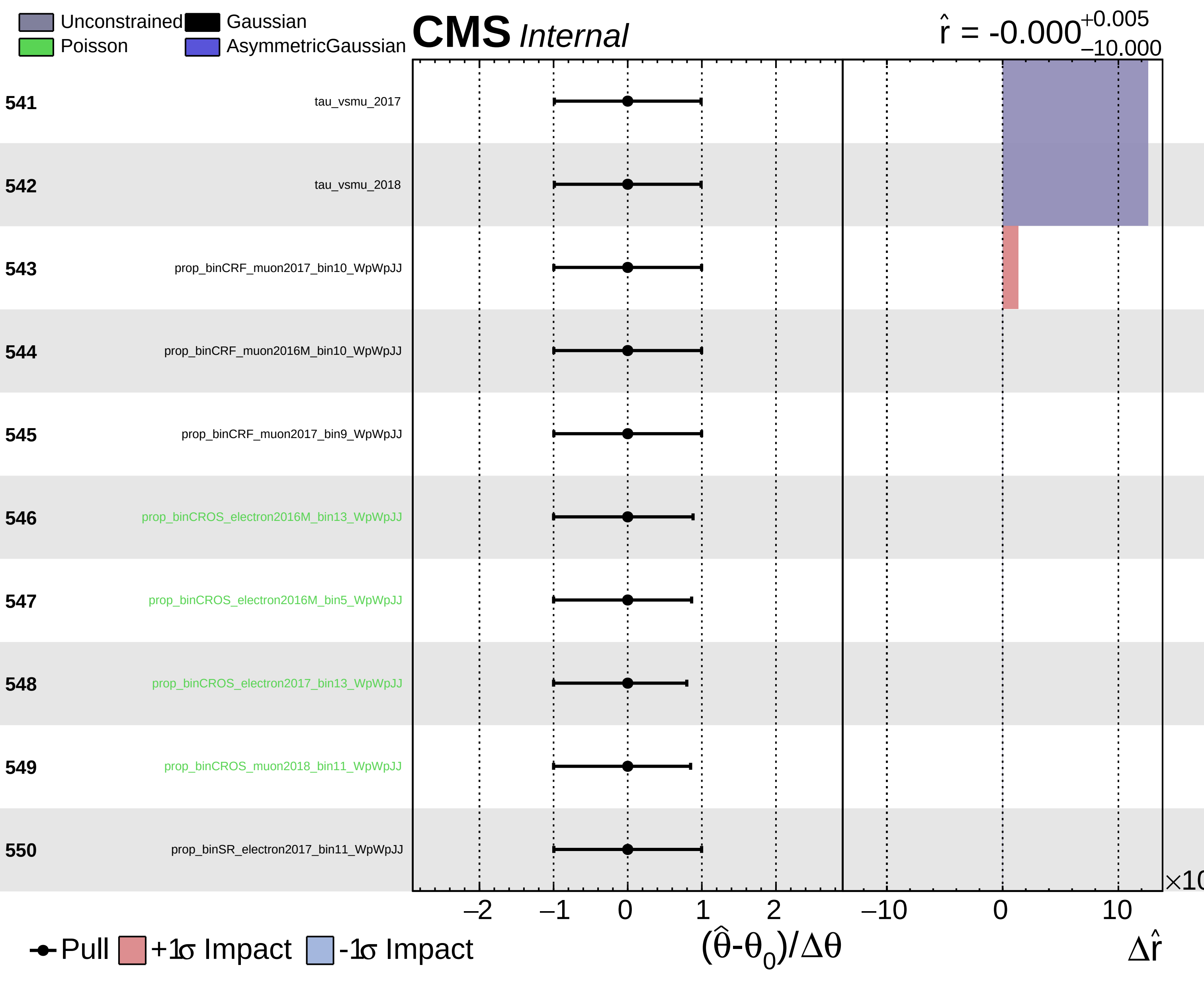

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.000^{+0.005}_{-10.000}$
421 prop_binCRF_muon2017_bin9_Triboson
422 prop_binCRF_electron2016M_bin8_Triboson
423 prop_binSR_electron2016M_bin13_WrongSign
424 prop_binCRF_muon2016M_bin10_Triboson
425 prop_binCRF_muon2016M_bin12
426 prop_binCROS_electron2017_bin12
427 prop_binSR_electron2016M_bin12_ZZtoLep
428 prop_binCROS_electron2018_bin10_VG
429 prop_binCROS_muon2017_bin8_TTTo2L2Nu
430 TTest_muon_2018
$1.00^{+0.05}_{-0.05}$
431 prop_binCROS_muon2018_bin13_WrongSign
432 prop_binCROS_electron2017_bin7_TTTo2L2Nu
433 prop_binCRTT_muon2016M_bin12
434 prop_binCROS_electron2017_bin9
435 prop_binCROS_electron2017_bin13_TTTo2L2Nu
436 prop_binCROS_muon2016M_bin12
437 prop_binCROS_muon2018_bin11_TTTo2L2Nu
438 prop_binCROS_electron2016M_bin13_WrongSign
439 prop_binCROS_electron2018_bin5
440 prop_binCROS_electron2018_bin8_TVX
441 prop_binCROS_electron2018_bin10_TTTo2L2Nu
442 prop_binSR_electron2016M_bin2_ZZtoLep
443 prop_binCROS_electron2016M_bin12_TTTo2L2Nu
444 prop_binCROS_electron2016M_bin13_TTTo2L2Nu
445 prop_binCROS_electron2018_bin13_TTTo2L2Nu
446 prop_binCROS_muon2018_bin13_TTTo2L2Nu
447 prop_binCROS_electron2017_bin13_Triboson
448 prop_binCRF_electron2018_bin11_Triboson
449 prop_binCROS_electron2017_bin7_WZ
450 prop_binCROS_electron2017_bin13_TVX
−2 −1 0 1 2
−0.04 −0.02 0 0.02 0.04
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.000 +0.005 −10.000
451 prop_binCROS_electron2016M_bin11_Fake
452 prop_binCROS_muon2018_bin11_WZ
453 prop_binSR_electron2016M_bin2_TVX
454 prop_binCROS_electron2018_bin9_TVX
455 prop_binCROS_muon2018_bin13_TVX
456 prop_binCROS_electron2016M_bin12_Triboson
457 prop_binCROS_electron2018_bin9_ZZtoLep
458 prop_binCROS_electron2016M_bin11_Triboson
459 prop_binSR_electron2016M_bin2_Triboson
460 prop_binCROS_electron2016M_bin11_TVX
461 prop_binCRF_electron2016M_bin9_TVX
462 prop_binCROS_electron2016M_bin13_Triboson
463 prop_binCROS_electron2018_bin9_TTTo2L2Nu
464 prop_binCROS_electron2016M_bin12_TVX
465 prop_binCROS_electron2017_bin7_VG
466 prop_binCROS_electron2018_bin13_Triboson
467 prop_binCROS_electron2017_bin7_Triboson
468 prop_binCROS_electron2017_bin7_TVX
469 prop_binCRF_electron2016M_bin9_Triboson
470 prop_binCROS_electron2016M_bin12_ZZtoLep
471 prop_binCROS_electron2018_bin10_TVX
472 prop_binCROS_electron2017_bin7_ZZtoLep
473 prop_binCROS_electron2018_bin10_Triboson
474 prop_binCROS_muon2016M_bin7_WZ
475 prop_binCROS_muon2016M_bin7_TVX
476 prop_binCROS_muon2016M_bin7_ZZtoLep
477 prop_binCRF_electron2016M_bin8_ZZtoLep
478 prop_binCROS_muon2018_bin11_Triboson
479 prop_binCROS_electron2016M_bin13_ZZtoLep
480 prop_binCROS_muon2018_bin11_TVX
−2 −1 0 1 2
−0.02 −0.01 0 0.01 0.02
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.000^{+0.005}_{-10.000}$
481 prop_binCROS_electron2018_bin13_TVX
482 prop_binSR_electron2016M_bin12_WrongSign
483 prop_binCROS_electron2018_bin8_ZZtoLep
484 prop_binCROS_electron2018_bin13_ZZtoLep
485 prop_binCROS_electron2018_bin10_ZZtoLep
486 prop_binCROS_muon2017_bin8_VG
487 prop_binSR_electron2017_bin13_ZZtoLep
488 prop_binCROS_muon2017_bin8_Triboson
489 prop_binCROS_muon2018_bin11_ZZtoLep
490 prop_binCROS_muon2017_bin8_ZZtoLep
491 prop_binCROS_electron2016M_bin5_Triboson
492 prop_binCROS_electron2016M_bin11_ZZtoLep
493 prop_binCROS_electron2016M_bin5_TVX
494 prop_binCROS_electron2017_bin6_Triboson
495 prop_binCROS_electron2016M_bin5_ZZtoLep
496 prop_binCROS_electron2017_bin6_ZZtoLep
497 lep_2016M
498 prop_binCRF_electron2017_bin13
499 prop_binCRF_electron2016M_bin9_ZZtoLep
500 prop_binCROS_electron2017_bin6_TTTo2L2Nu
501 lep_2018
502 prop_binCROS_electron2016M_bin11_TTTo2L2Nu
503 FES
504 prop_binCROS_muon2018_bin13_Triboson
505 prop_binCROS_muon2016M_bin7_Triboson
506 QCDScale_sig
507 VBS_2017
508 VBS_2018
509 prop_binCROS_electron2018_bin10_WpWpJJ
510 prop_binSR_electron2016M_bin12_WpWpJJ
−2 −1 0 1 2
−0.01 0 0.01
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.000 +0.005 −10.000
511 prop_binSR_electron2016M_bin8_WpWpJJ
512 prop_binCRF_electron2016M_bin9_WpWpJJ
513 prop_binCROS_electron2016M_bin12_WpWpJJ
514 prop_binCROS_electron2017_bin6_WpWpJJ
515 prop_binCRF_electron2018_bin11_WpWpJJ
516 prop_binCRF_electron2017_bin9_WpWpJJ
517 prop_binSR_muon2017_bin11_WpWpJJ
518 prop_binCROS_electron2017_bin7_WpWpJJ
519 prop_binCROS_muon2016M_bin7_WpWpJJ
520 prop_binCROS_electron2017_bin13_WrongSign
521 prop_binCROS_electron2018_bin8_Triboson
522 prop_binCROS_electron2018_bin9_Triboson
523 prop_binCROS_electron2017_bin6_TVX
524 VBS_2016M
525 prop_binCROS_electron2018_bin13_WpWpJJ
526 prop_binCROS_electron2016M_bin11_WpWpJJ
527 prop_binSR_electron2016M_bin2_WpWpJJ
528 prop_binCROS_electron2018_bin8_WpWpJJ
529 prop_binSR_electron2018_bin13_WpWpJJ
530 prop_binSR_muon2016M_bin8_WpWpJJ
531 prop_binCROS_muon2017_bin8_TVX
532 prop_binSR_electron2016M_bin13_WpWpJJ
533 prop_binCROS_electron2018_bin9_WpWpJJ
534 prop_binSR_electron2017_bin13_WpWpJJ
535 prop_binCRF_electron2016M_bin8_WpWpJJ
536 QCDScale_WZ
537 tau_vsele_2016M
538 tau_vsele_2017
539 tau_vsele_2018
540 tau_vsmu_2016M
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.000^{+0.005}_{-10.000}$
541
tau_vsmu_2017
542
tau_vsmu_2018
543
prop_binCRF_muon2017_bin10_WpWpJJ
544
prop_binCRF_muon2016M_bin10_WpWpJJ
545
prop_binCRF_muon2017_bin9_WpWpJJ
546
prop_binCROS_electron2016M_bin13_WpWpJJ
547
prop_binCROS_electron2016M_bin5_WpWpJJ
548
prop_binCROS_electron2017_bin13_WpWpJJ
549
prop_binCROS_muon2018_bin11_WpWpJJ
550
prop_binSR_electron2017_bin11_WpWpJJ
−2 −1 0 1 2
−10 0 10
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$